VESDA Pipe Network Design Guide

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Scope

The VESDA Pipe Network Design Guide will help you with the design specifications and management of VESDA systems.

While the ASPIRE software helps you design an effective pipe network, this guide will cover the principles of good design and aims to assist you in producing the optimum design for a site.

Document Conventions

The following typographic conventions are used in this document:

Convention	Description
Bold	**Used to denote:** emphasis. Used for names of menus, menu options, toolbar buttons
Italics	**Used to denote:** references to other parts of this document or other documents. Used for the result of an action.

The following icons are used in this document:

Convention	Description
	Caution: This icon is used to indicate that there is a danger to equipment. The danger could be loss of data, physical damage, or permanent corruption of configuration details.
	Warning: This icon is used to indicate that there is a danger of electric shock. This may lead to death or permanent injury.
	Warning: This icon is used to indicate that there is a danger of inhaling dangerous substances. This may lead to death or permanent injury.

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Codes and Standards Information for Air Sampling Smoke Detection

We strongly recommend that this document is read in conjunction with the appropriate local codes and standards for smoke detection and electrical connections. This document contains generic product information and some sections may not comply with all local codes and standards. In these cases, the local codes and standards must take precedence. The information below was correct at time of printing but may now be out of date, check with your local codes, standards and listings for the current restrictions.

FCC Compliance Statement

This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instruction, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, the user is encouraged to try to correct the interference by one or more of the following measures; re-orientate or relocate the receiving antenna, increase the separation between the equipment and receiver, connect the equipment to a power outlet which is on a different power circuit to the receiver or consult the dealer or an experienced radio/television technician for help.

FDA

This VESDA product incorporates a laser device and is classified as a Class 1 laser product that complies with FDA regulations 21 CFR 1040.10. The laser is housed in a sealed detector chamber and contains no serviceable parts. The laser emits invisible light and can be hazardous if viewed with the naked eye. Under no circumstances should the detector chamber be opened.

FM Hazardous Applications

3611 Hazardous Approval Warning: Exposure to some chemicals may degrade the sealing of relays used on the detector. Relays used on the detector are marked “TX2-5V”, “G6S-2-5V” or “EC2-5NU”.

VESDA detectors must not be connected or disconnected to a PC while the equipment is powered in an FM Division 2 hazardous (classified) location (defined by FM 3611).

ONORM F3014

ONORM F3014, transport times for all tubes (including capillaries) must not exceed 60 seconds from any hole. This means that the pre-designed pipe networks that include capillaries cannot be used.

AS 7240.20

The performance of this product is dependent upon the configuration of the pipe network. Any extensions or modifications to the pipe network may cause the product to stop working correctly. You must check that ASPIRE approves alterations before making any changes. ASPIRE is available from your VESDA distributor.

AS1851.1 2005

Maintenance Standards. Wherever this document and the AS1851.1 differ, AS1851.1 should be followed in preference to this document.

European Installations

The product must use a power supply conforming to EN54: Part 4.

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1 Introduction to Good Design

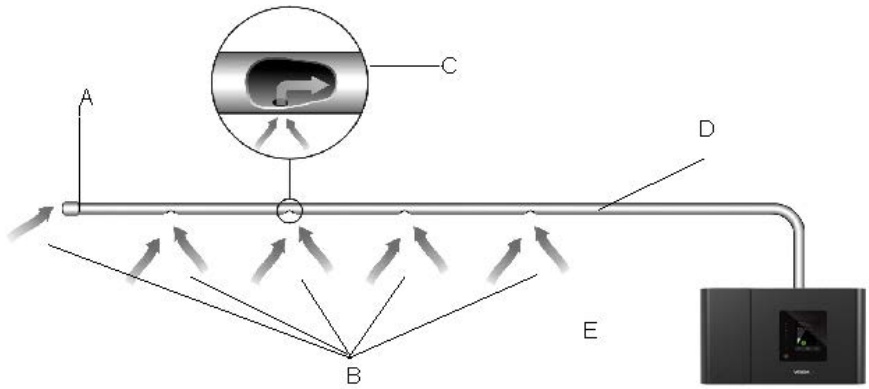
We recommend following the process outlined below when developing site specifications to ensure an efficient, compliant, and optimized design.

1. Ensure you have a clear understanding of all applicable local codes, regulations, and standards relevant to the site.
2. Attend accredited training programs, as they significantly support the development of an optimal site design.
3. Collect comprehensive and accurate information about the site. This guide will assist you in doing this correctly.
4. Use the ASPIRE software to test and optimize the design.

Exceeding the guidelines specified in this guide is acceptable, provided the design is validated and confirmed by the ASPIRE software.

1.1 Introduction to Pipe Network Design

The VESDA system collects air samples through sampling holes on a network of pipes. The airflow within a protected area carries the air samples to the sampling holes. Conventional smoke detectors wait for the smoke to migrate through the detector, while an aspirating smoke detection system actively draws air samples into the sampling pipes. These samples are transported through the pipe network to the VESDA detector.



Legend	
A	End Cap with hole
B	Air samples
C	Detail showing airflow entering a sampling hole
D	Air sampling pipe
E	Detector

Figure 1-1: VESDA Air Sampling System

1.2 Before You Start

To design an efficient pipe network you will need to:

- Have knowledge of your local codes and standards.
- Undergo accredited training in pipe network design and the ASPIRE software.
- Have access to a floor plan for the protected area. The floor plan must include details of existing or proposed fixtures, fittings and equipment.
- Have information about the purpose of the area to be protected. Warehouses, cold stores, computer rooms and other applications may require special considerations.
- Determine the protection level required.

2 Designing a Pipe Network

To design an effective pipe network, it is suggested that the following steps are taken. The order in which these are completed may differ with each project.

- Collect site information
- Define VESDA Addresses (also known as VESDA Zones)
- Select appropriate sampling method(s)
- Select appropriate detector
- Plan and map a pipe network
- Calculate design performance using the ASPIRE software
- Record the details of the optimum design

3 Collecting Site Information

As a first step it is essential to collect information about the site to be protected. For an existing site this may include a site survey prior to designing a pipe network. Most of the information required for an effective pipe network design can be determined by a site visit. For sites yet to be constructed or where a site visit is not possible, a site plan can be used to aid pipe design. The information required through a site survey includes:

- Site layout and measurements
- Regulatory requirements
- Air flow within the protected area
- The ambient conditions within the site
- The purpose for which the site is to be used
- Construction of the site (beams, beam pockets, and pipe obstructions)
- The anticipated influence of the external environment factors on the protected area

3.1 Site Layout and Measurements

Before designing can commence, a good knowledge of the site layout is necessary. A plan showing measurements of the area to be protected assists with the planning of fire zones and VESDA Addresses. The site layout also shows areas designated for different uses and obstacles to free flow of air (partitions, air curtains, etc.). Areas requiring special protection, and the location of plant, machinery, equipment, cabinet layout, and the rack layout in warehouses, must be identified on the site plan.

3.2 Regulatory Requirements

The designer determines the local codes and standards that are applicable to the site. These need to be considered when creating fire zones, VESDA Addresses, and the pipe network design.

Note: Local codes and standards have precedence over any VESDA recommendations. Where the parameters set by the VESDA product are not the same as those set by the local codes and standards, the local codes and standards should be adopted.

3.3 Air Flow

When designing a pipe network, it is essential to identify and understand the natural airflow patterns within the area to be protected. Consideration must be given to any existing or proposed mechanical ventilation systems, air curtains, containment aisles, roller doors, or partitions that may affect the free flow of air. Where possible, conduct smoke tests to determine the air flow. The stability or fluctuation in air flow conditions must be factored into the pipe network design to ensure optimal performance. Accurate interpretation of airflow patterns is critical, as it directly influences the pipe network’s ability to detect a fire at its earliest stages.

3.4 Ambient Conditions

Where possible, the ambient conditions existing within the different areas of a site should be identified. It is likely that conditions within a site differ from one area to the other. The efficiency of the VESDA system is dependent upon the accurate determination of the ambient conditions of the protected area.

3.5 Site Purpose

The purpose of the site as well as the protected area need to be considered when designing a pipe network. Typically, a site may comprise of an office area, warehouse, a factory, computer room, and cafeteria, each requiring special consideration when designing a pipe network. Certain manufacturing and processing areas may produce smoke, dust, steam, flame or heat. Consideration must be given to these conditions in the Pipe Network Design Guide.

3.6 Site Construction

The designer should consider:

- The material used in the construction of the site. (It is easier to run pipes through plaster than concrete walls)
- The internal design and the material used for internal surfaces, decoration and furniture
- The types of rooms or areas to be protected (high ceilings, high air exchange rate)
- The existence and use of ceiling and floor voids (beams and beam pockets)
- Obstructions to pipe layout or the free movement of air
- Placement of equipment requiring special protection (object detection)
- Location of mechanical ventilators, Air Handling Units (AHUs), return air ducts, and supply and exhaust air systems
- The beam layout, beam pockets and other 'odd' construction spaces

3.7 Surrounding Environment

Attention should be paid to the environment surrounding the site. If the site is situated in an area of high pollution levels, the VESDA zone(s) subjected to frequent exposure to external environment may record unexpected increase in the background level of smoke. To compensate for this, reference detectors may be required. Typically certain areas within the site like warehouses and loading bay areas are normally subject to external environmental influences and require special attention during the design of the pipe network. Refer to Section 4.10 on page 22 for further information.

4 Air Sampling Methods

Site requirements and conditions determine the best sampling method. Local codes and standards also influence the selection of the sampling methods used. Air from an Address (Zone) is drawn through sampling holes in the pipe network. The pipe network transports the sampled air to the detector. The position and spacing of sampling holes is dependent upon the pipe network design. Refer to section 4.11 on page 23 for examples of the sampling methods that can be used for various applications.

There are three basic air sampling methods (as described below). Typically, a site will use more than one.

4.1 Standard Pipe Sampling

These guidelines are used for all pipe network designs. In addition, guidelines specific to different sampling methods for different types of applications may also apply. Refer to Chapter 7 for further information.

1. Parameters and values specified in your local codes and standards have precedence over anything suggested in Xtralis documentation.
2. We recommend smooth bore pipe with 25 mm Outer Diameter (OD) and 3/4 inch IPS Internal Diameter (ID). The preferred CPVC, PVC, ABS or UPVC pipe ID is 21 mm (0.874 in).
3. Maintain at least 500 mm (20 inches) of straight pipe before the pipe terminates at the detector.
4. Use bend or elbow connectors to change the direction of the pipe. Bends maintain better airflow than elbows.
5. All sampling pipes should be fitted with an end cap. Use the ASPIRE software to assist in determining the size of the hole in the end cap.
6. For optimum detector performance you should keep pipe runs to a similar length.
7. For optimum detector performance you should keep a similar number of holes in each pipe.
8. It is better to use multiple pipes of shorter total length, than a single pipe of longer length.

A simple grid layout can be used to suit most rectangular spaces. Refer to Section 6.1 on page 30 for further information.

4.2 Capillary Tube Sampling

Capillary tubes are used to sample air some distance away from the sampling pipe and in enclosed areas like cabinets. The guidelines for capillary sampling are:

1. Whereeeee possible, use capillaries of the same length.
2. The maximum capillary tube ID should be 5 mm (3/8 in.).
3. Maintain balance within the pipe network by using the ASPIRE software to determine the size of the sampling hole required.
4. Maximum length of the Capillary tube shall be 8 m (26 ft).
5. The positioning of the sampling hole for in-cabinet sampling is dependant on air flow conditions. In most instances, the sampling hole is positioned close to the top of the interior of the cabinet. Refer to Figure 4-16 on page 16 for further information.

If you wish to use capillary sizes outside these guidelines, please use the ASPIRE software to verify that the resulting design remains acceptable.

4.3 Open Area Protection

4.3.1 On-ceiling Sampling

Typically, the pipe network is suspended approximately 25 mm to 300 mm (1 in to 1 ft) below the ceiling in the area to be protected.

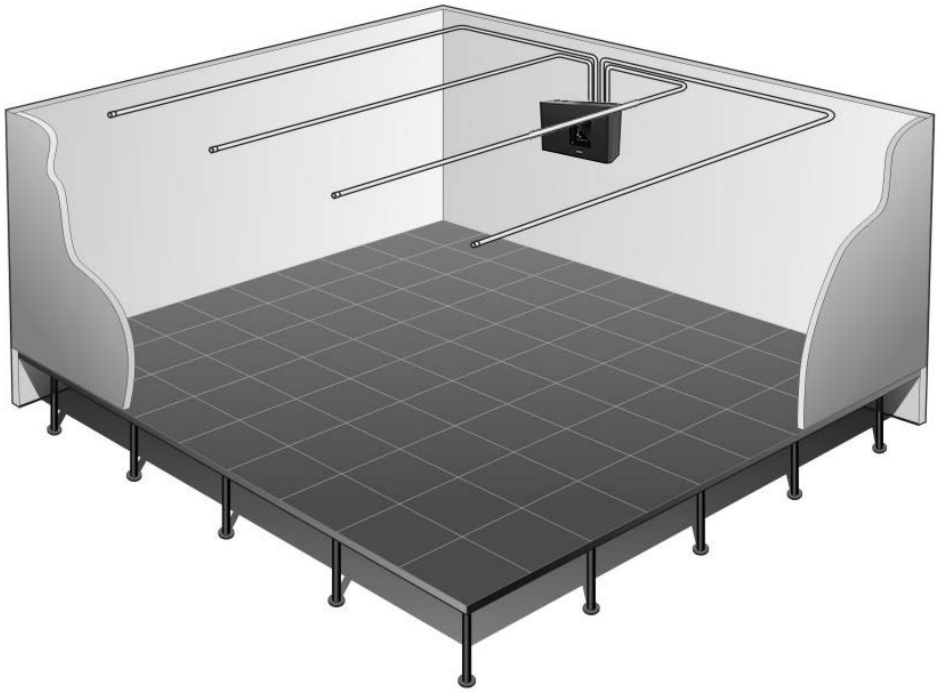
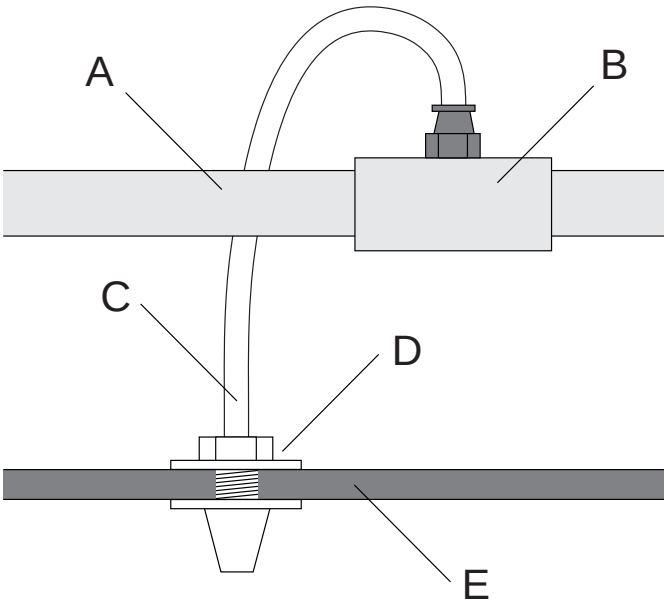


Figure 4-1: A typical on-ceiling installation

4.3.2 Concealed Sampling Pipe Network

Capillary tube sampling is ideal where the sampling pipe needs to be concealed for aesthetic or security reasons. The main pipe network is installed in the ceiling void with capillary tubes branching off at required intervals to penetrate the ceiling panels or tiles.

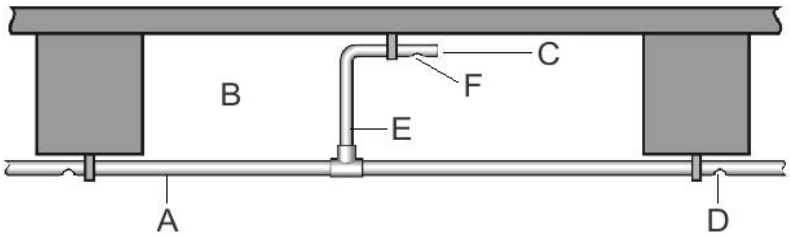


Legend	
A	Sampling pipe
B	Tee adaptor
C	Capillary tube
D	Miniature sampling point
E	Ceiling

Figure 4-2: Detail of concealed sampling pipe network with conical point fittings

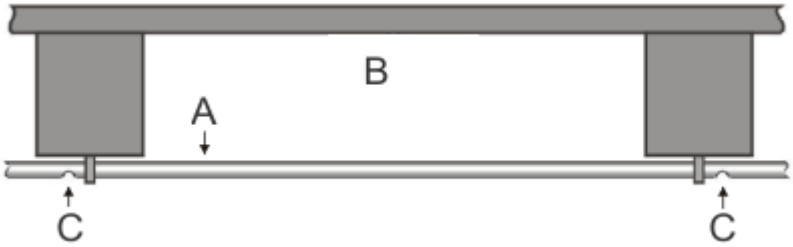
4.3.3 Inter-Beam Sampling

Beam pockets are created between large ceiling beams. The protection of beam structure ceilings should follow local codes and standards requirements or through an alternate solution by a Performance-Based Design approach. Where protection of the inside of beam pocket is required, a rigid pipe may extend vertically from the sampling pipe upwards into the space between the beams (E). At the top end the direction of the pipe is changed to horizontal. The end of the horizontal sampling pipe is capped with a sealed end cap which may, or may not have a sample hole in it (C). A sampling hole is drilled just before the end cap (D). Underside beam sampling and/or inside beam sampling can be used. Refer to Figure 4-12 on page 13.



Legend	
A	Sampling pipe
B	Beam pockets
C	End cap
D	Sampling hole
E	Vertical sampling pipe
F	Underside sampling hole

Figure 4-3: Inter-beam sampling using a walking stick



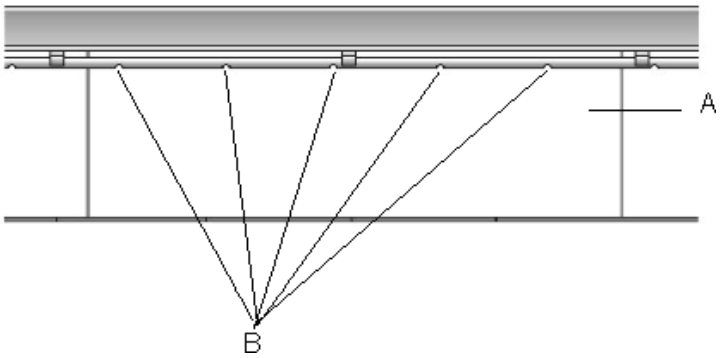
Legend	
A	Sampling pipe
B	Beam pockets
C	Sampling Holes

Figure 4-4: Inter-beam sampling

4.4 Return Area Protection

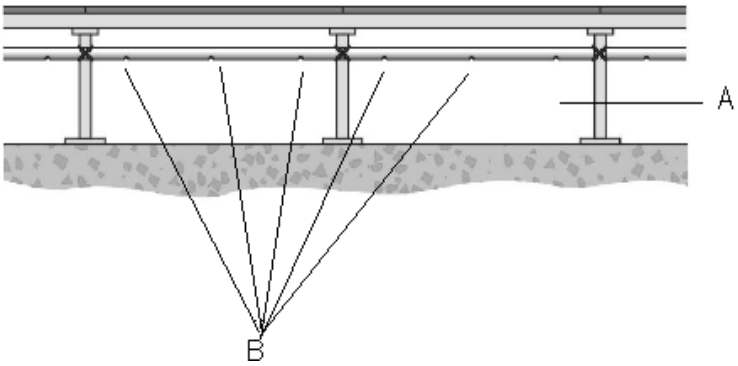
4.4.1 In-Ceiling or Floor Void Sampling

Some applications use ceiling and under floor voids as return air plenums (ducts). The pipe network is designed to monitor the air flowing through the return air plenums. In-ceiling or floor void sampling is also used to monitor any cabling and equipment that may be installed in the ceiling and floor voids. In instances where the airflow is perpendicular to the pipe then it is recommended that sampling holes have a 30° orientation to the airflow. Refer to Figure 4-12 on page 13.



Legend	
A	Ceiling void
B	Sampling holes

Figure 4-5: Ceiling void sampling



Legend	
A	Floor void
B	Sampling holes

Figure 4-6: Floor void sampling

4.4.2 Return Air Sampling

Smoke tends to travel with any mechanically generated air flow. Correctly positioning sampling holes in a pipe network across the return air grille of an AHU or an exhaust ventilation system, ensures that any smoke is detected at the earliest stage. Air samples from inside a duct carrying the exhaust air may also be collected, refer to Figure 4-8 for details. It is recommended that the sampling holes face between 20 and 45 degrees from the direction of the greatest airflow. Refer to Figure 4-7 for further information.

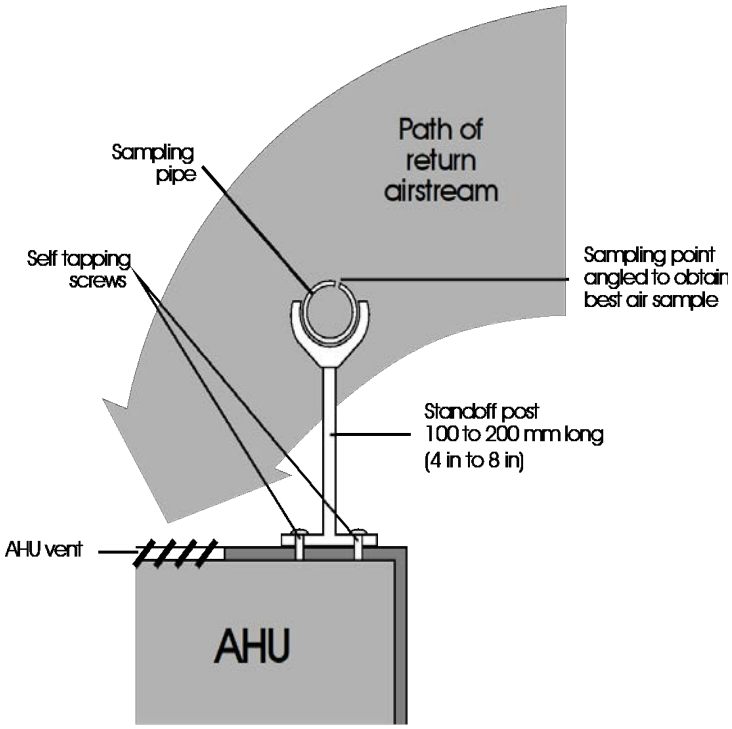


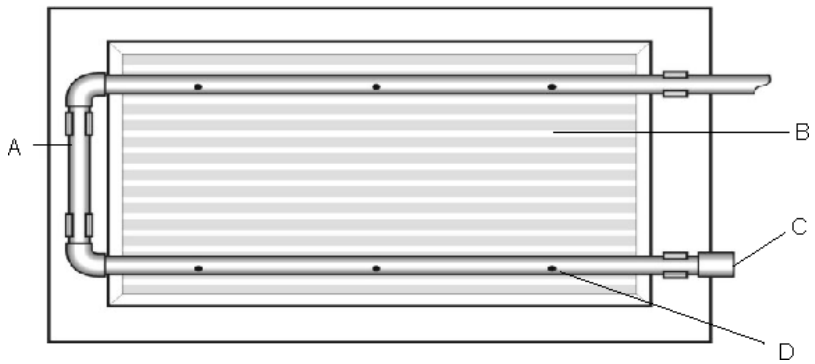
Figure 4-7: Cross section of pipe position on a return air grille

4.4.3 Return Air Grille Sampling

The sampling pipe is placed over the return air grille of a duct or an AHU. Figure 4-8 illustrates pipe mounting over the return air grille. The number of AHUs that can be covered by different detectors is dependent on the suction power of the AHU and also on the size of the return air grille.

Recommendations:

- VESDA-E VEP or VESDA-E VES detector should not monitor more than four AHUs.
- VESDA-E VEP-1P should not monitor more than two AHUs.
- VESDA VLF should not monitor more than one AHU.
- The detectors should be in close proximity to the AHUs.
- Transport time should be less than 60 seconds.
- Sampling hole sensitivities should be maintained.



Legend	
A	Sampling pipe
B	Return grill
C	End cap without hole
D	Sampling hole

Figure 4-8: Return air sampling over a return air grille

Note: Socket unions are also used where it is important to have correct orientation of sampling holes.

4.4.4 Hot/Cold Aisle Sampling

In hot/cold aisle systems, the cabinets are adjoined into a series of rows, resting on a raised floor. The fronts of the racks face each other and become cold aisles, due to the front-to-back heat dissipation of rack servers. Hot/cold aisle protocol can be implemented with underfloor, overhead, horizontal displacement, or local air distribution systems.

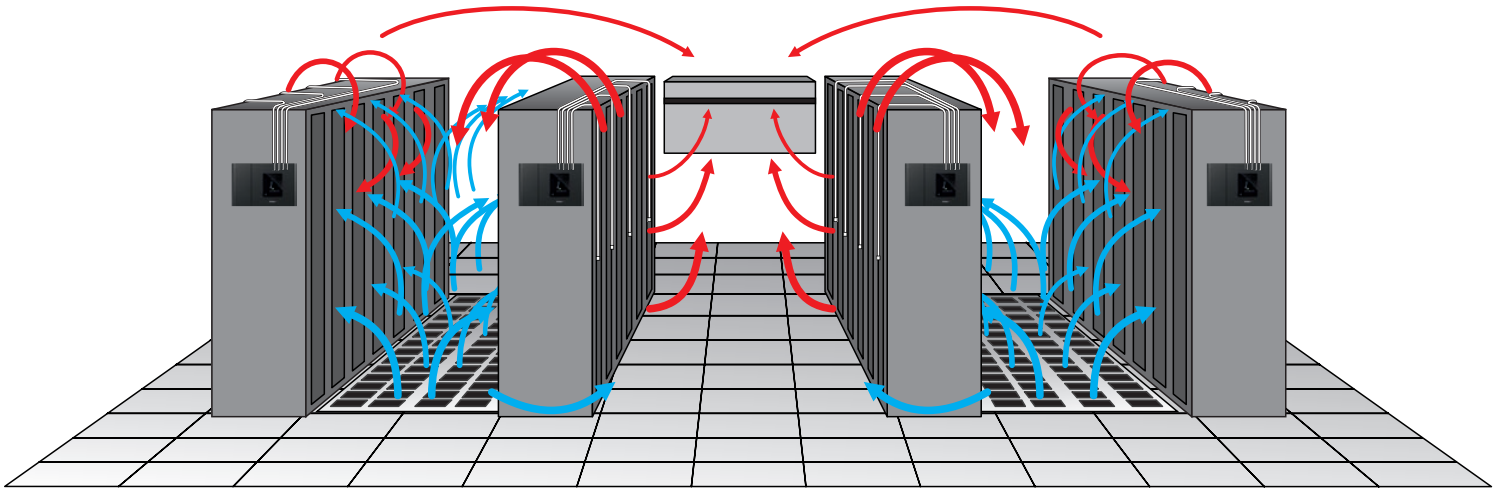


Figure 4-9: Hot/Cold Aisle System

- **Primary Sampling:** rear of racks, air-return grills (CRAC, Ceiling)
- **Secondary Sampling:** ceiling.

4.4.5 Hot Aisle Containment Sampling

In hot aisle containment systems, racks are positioned in rows (back to back). The hot aisle in between racks will be covered on the top and at the end of the rows and ducted back to the air conditioning unit (Figure 4-10). The result is a full separation between supply and return air.

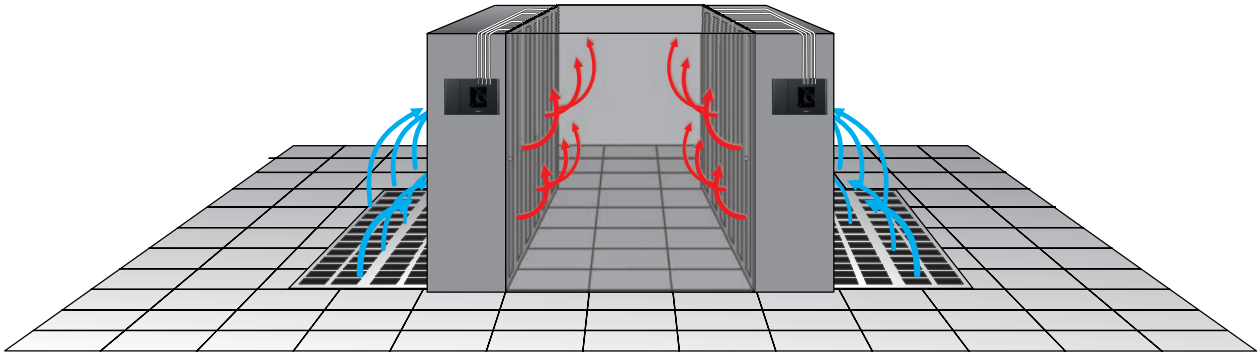


Figure 4-10: Hot aisle containment sampling

4.4.6 Cold Aisle Containment Sampling

In cold aisle containment, racks are positioned in rows (front to front). The cold aisle in between racks will be covered on the top and at the end of the rows. The result of this is a full separation between supply and return air. Cold air will be supplied through a raised floor into the contained cold aisle; hot return air is exhausted from the racks into the room.

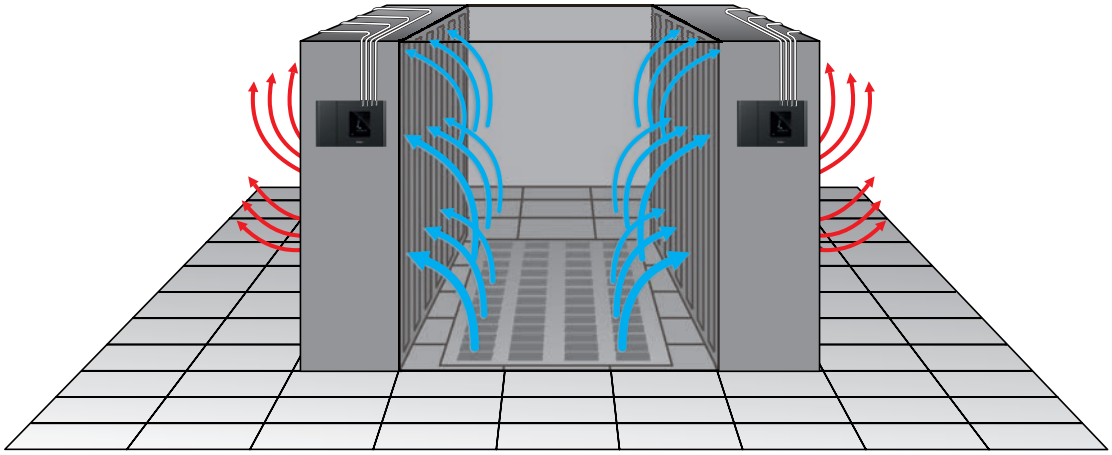
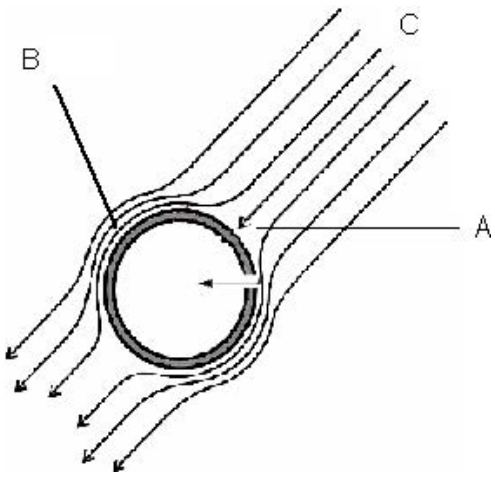


Figure 4-11: Cold aisle containment sampling

4.4.7 Hole Orientation

Industry experience shows that the pipework can be fine - tuned by locating the sampling holes 30 degrees away from the airflow path. Sampling can be improved by avoiding the high and low velocity areas. One common problem with this model is that the detector needs to work in all possible situations. If you are installing pipework into a high airflow area keep in mind what will happen if/when the AHUs are turned off.



Legend	
A	Low velocity (high static pressure) area
B	High velocity (low static pressure) area
C	Airflow streamlines

Figure 4-12: Sampling hole orientation

Note: Socket unions are also used where it is important to have correct orientation of sampling holes.

Guidelines for return air sampling include:

- Place the sample pipe (probe) in the path of the greatest airflow.
- More than one sampling hole may be required for large grilles. NFPA 76 recommendations specify that each sampling hole can cover a maximum of 0.4 m² (4 sq.ft).
- Care should be taken to keep the number of bends to a minimum.
- We recommend that sampling holes should be angled at 30 degrees to the airflow.
- In high velocity air flows, it is advisable to use standoff fittings to keep the sampling pipes at least 25 mm to 200 mm (1 to 8 in.) in front of the grille. Installing any closer to the grille will put the sample point in an area of negative air pressure.
- The pipe will require capping with an end cap without a hole.

- Where maintenance requires the removal of the sampling pipes on a regular basis, the pipe network design should provide for use of socket unions to ensure correct orientation of sampling holes on re-connection.

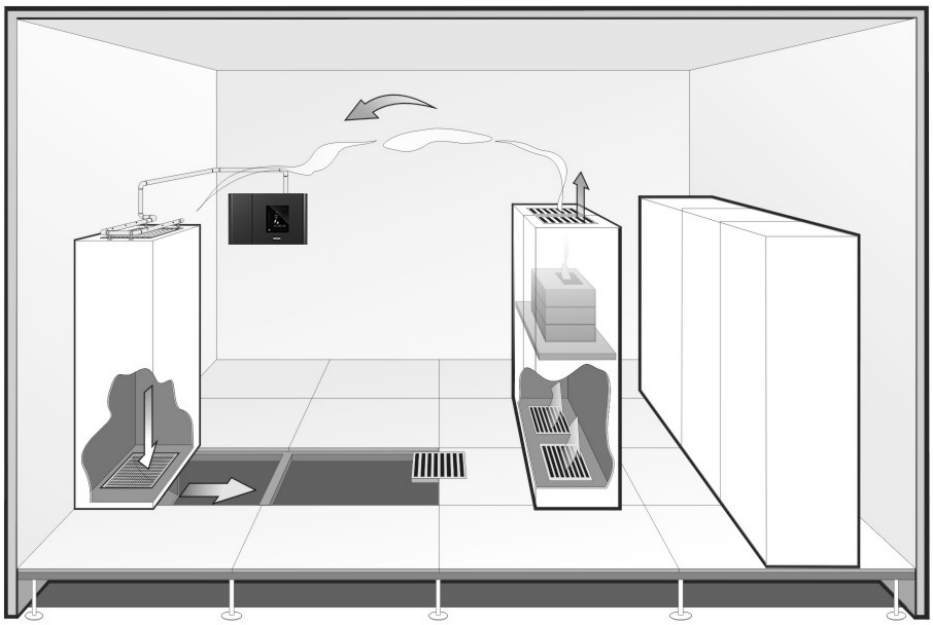


Figure 4-13: Typical return air sampling application

4.5 Object Protection

4.5.1 Above Cabinet Sampling

The sampling pipe is installed directly over the cabinets to be protected. Sampling holes are placed over the cabinet ventilation grille. The holes are drilled so that they face into the air stream from the cabinet. Each monitored cabinet must have at least one sampling hole. The exhaust fan can cause sampling problems due to the high airflow around the sample pipe.

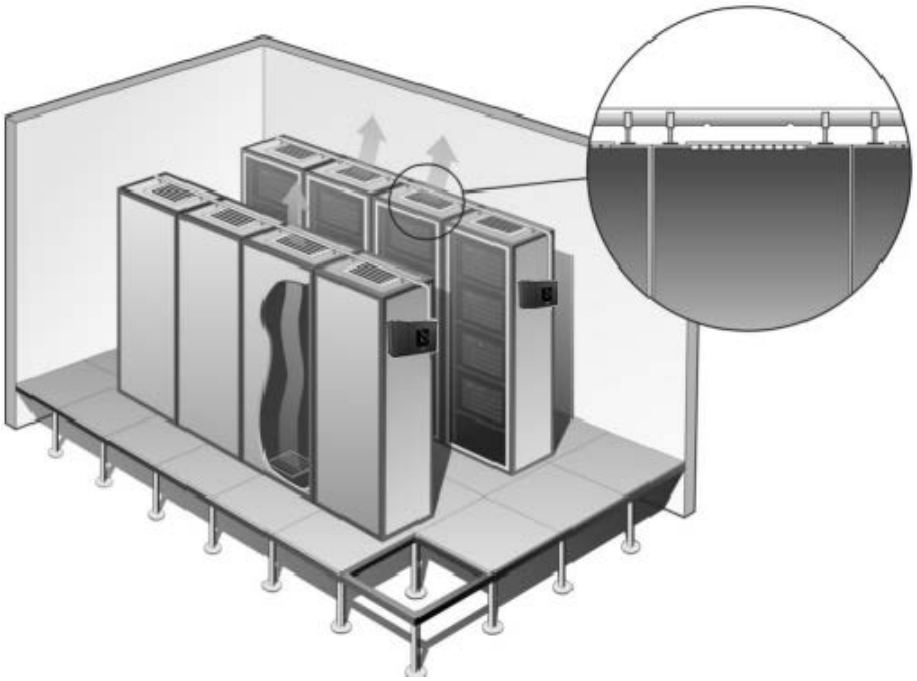
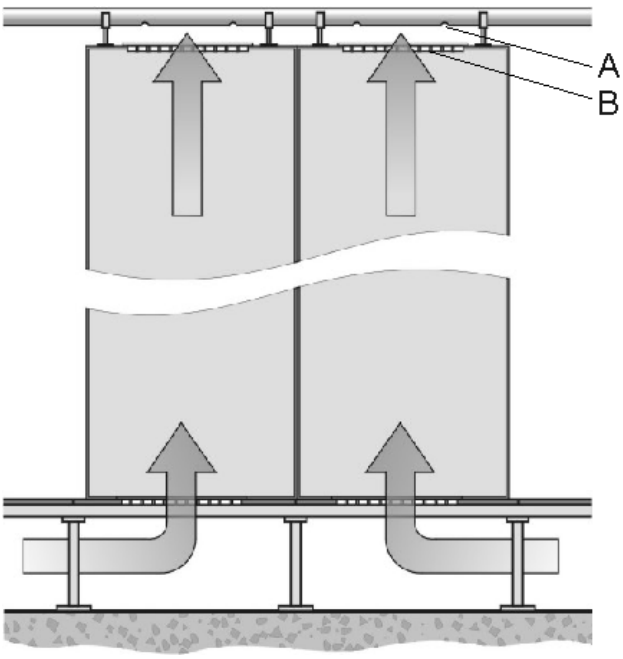


Figure 4-14: Exhaust grille sampling on a bank of cabinets

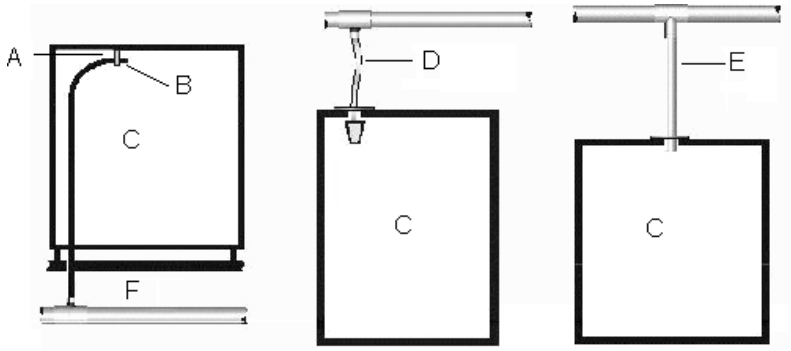


Legend	
A	Sampling hole
B	Air grille

Figure 4-15: Cabinet sampling for convection cooled cabinet

4.5.2 In-Cabinet Sampling

Capillary tubes are used for enclosed in-cabinet sampling. A flexible capillary tube of a maximum length of 8 m (26 ft) is connected to the main pipe network using tee connection with reducing adaptors. A variation to the capillary sampling system uses a drop pipe. A 12.5 mm (0.5 inch) ID rigid pipe is connected to the sampling pipe via a tee connection.



Legend	
A	Retainer clips
B	Sampling hole
C	Equipment cabinets
D	Capillary tube
E	Rigid drop pipe
F	Underfloor void

Figure 4-16: Illustration of in-cabinet sampling using capillaries and drop pipes

Note: Care must be taken when installing sample points on the top of cabinets with extractor fans. These fans may create low air pressure in the cabinet which may stop any air samples being able to enter the sampling point.

4.5.3 Cable Trays

Cable trays are used to support the distribution of electric cables between racks and equipment, and are normally suspended from ceiling through cable hangers.

The system design will normally provide ceiling protection, but adequate fire protection in rooms with cable trays will depend on the cable tray area relative to the room area and density of the cables on the tray.

A performance-based design evaluates the distribution of smoke within a room containing cable trays and whether the sampling holes can be confined to the ceiling or need to be extended below the cable trays using drop down pipes.

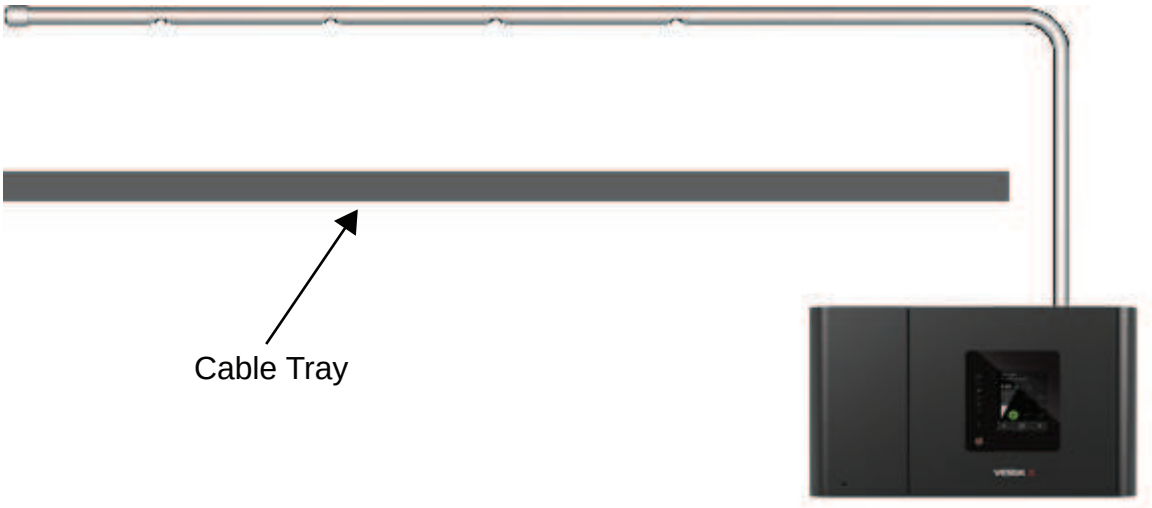


Figure 4-17: Ceiling sampling over a cable tray

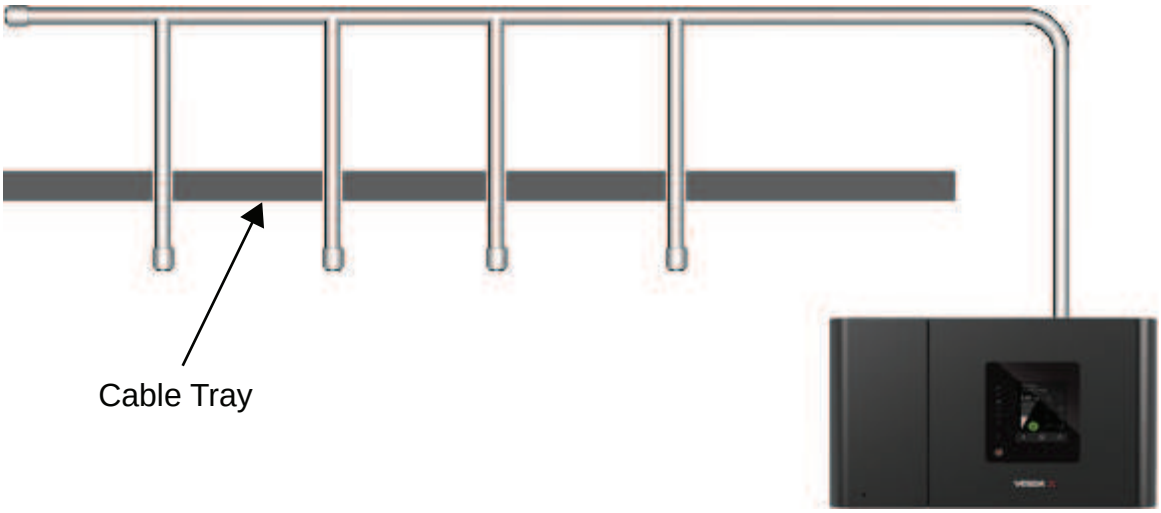


Figure 4-18: Ceiling and drop-down pipe sampling

Note: Ceiling and under cable tray sampling holes should be arranged in a staggered configuration.

4.6 In-Duct Sampling

In a fire event, ventilation duct systems can convey smoke; usually hot toxic gases and flames from one area to another. Duct systems may also supply air to aid combustion in the fire location. Therefore, the effective management of smoke control, such as dampers and shutters to contain smoke spread and fire growth is essential for life safety.

VESDA VLF models VLF-500 and VLF-250 or VESDA-E VEP-1P is recommended for duct applications. Xtralis has conducted extensive testing in duct environments to determine the optimum installation parameters.

Key Design Considerations:

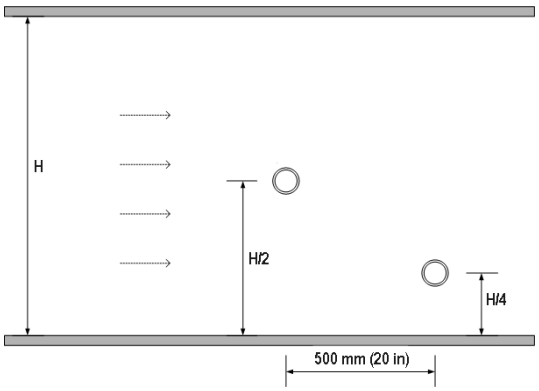
Sampling tubes should be oriented to overcome thermal stratification due to buoyancy of the smoke in the upper half of the duct. This condition occurs where duct velocities are low, buoyancy exceeds flow inertia, or the detector is installed close to the fire compartment. This condition occurs in the upper half of the duct. The effects of differential buoyancy are overcome by vertical orientation of sampling tubes. In a single fire compartment, the detector sampling tube is located in the upper half of the duct to minimize the effects of thermal stratification.

Where a detector is installed on a duct serving a single fire compartment, where the buoyancy exceeds the flow inertia of the air in the duct and the sampling tube cannot be oriented vertically, then the effects of thermal stratification can be minimized by locating the detector sampling tube in the upper half of the duct.

- The inlet pipe must be inserted at a distance between six to ten duct widths (width is considered the longest side for oblong ducts) or diameters from any disturbances to the flow generated by sharp bends, plenums, nozzles, branch connections, etc.
- The inlet and exhaust pipes must have the same length. They must be sealed at the far end with an end cap.
- The holes on the inlet and exhaust pipes should be facing the airflow as shown in Figure 4-20. Holes with the same orientation eliminate unwanted flow faults associated with cyclical operation, maintenance or power failure of the duct system. However, in some industrial applications where the quality of air inside the duct is poor, it is recommended to face all holes on the inlet and exhaust pipes downstream (i.e. 180° to incoming airflow).
- The pipes should always be supported at the duct walls by using fittings such as a rubber grommet.
- Appropriate sealant must be used to ensure an airtight seal.
- Make sure that the sampling hole at either end is at least 50 mm (2 in) from the duct walls.
- The exhaust pipe must have 4 x Ø10 mm (13/32 in) holes. The holes should be concentrated in the middle of the duct's width and spaced accordingly.

Small Duct: Width < 1 m (3 ft)

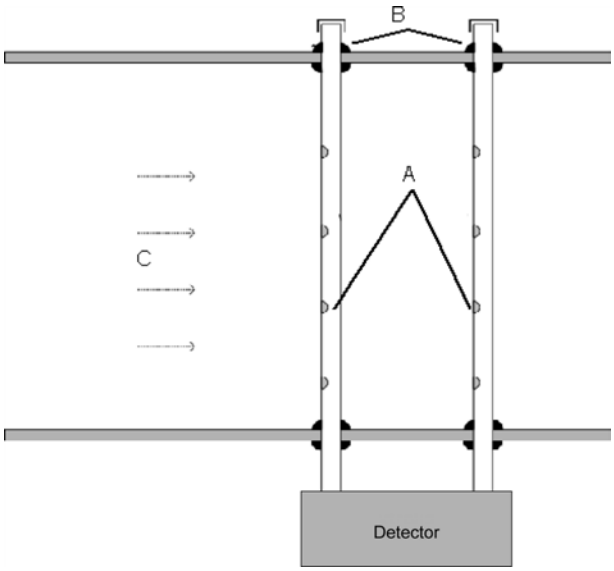
Figure 4-19 shows a side view of a duct section with the insertion positions for the inlet and exhaust pipes. The inlet pipe is to be installed in the middle of the duct height (H) or diameter. The exhaust pipe should be inserted approximately 0.5 m (1.64 ft) further downstream at a quarter of the height of duct.



Legend	
A	Exhaust pipe
B	Inlet pipe
C	Air flow

Figure 4-19: Small duct sampling - side view

The details of the number and size of the holes to be used can be found in Table 4-1.



Legend	
A	Holes with same orientation
B	Rubber grommet
C	Air flow

Figure 4-20: Small duct sampling - top view

For small ducts, holes are nominally spaced each 200 mm (8 in).

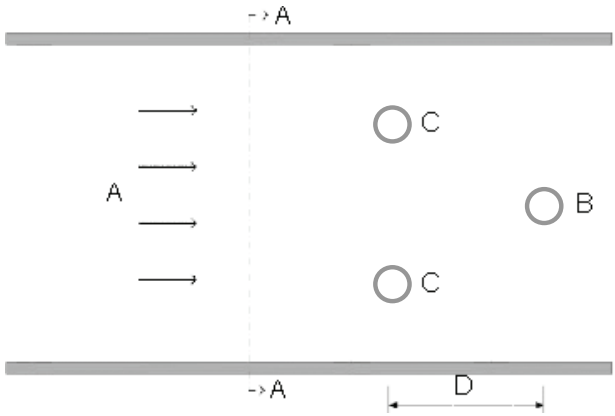
Table 4-1: Hole size for a small duct

Duct width	No. of holes	Hole Ø	Nominal pipe flow rate (L/min.) (cfm)
300 mm (12 in)	2	6 mm 15/64 in	39.0 L/min. (1.4 cfm)
500 mm (20 in)	3	5 mm 13/64 in	40.7 L/min. (1.4 cfm)
700 mm (28 in)	4	4 mm 5/32 in	35.6 L/min. (1.26 cfm)
900 mm (36 in)	5	4 mm 5/32 in	42.8 L/min. (1.51 cfm)

Large Ducts: Width 1 - 2 m (3 - 7 ft)

For large ducts, the inlet pipe is recommended to have two branches. Figure 4-22 shows a side view of a duct section with the relative insertion positions for the inlet and exhaust pipes. Both inlet branches enter at a quarter of the height of the duct from the top and bottom where H is the height of the duct.

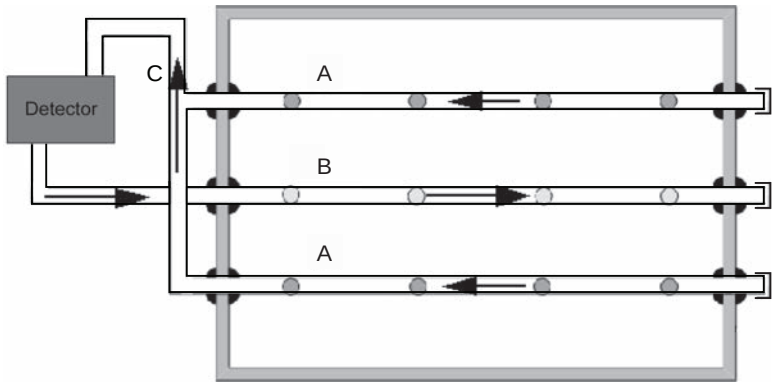
The exhaust pipe should be inserted approximately 0.5 m (1.64 ft) further downstream in the middle of the height of the duct.



Legend	
A	Duct flow
B	Exhaust pipe H/2
C	Inlet pipe H/4
D	500 mm (20 in.)

Figure 4-21: Large duct sampling - side view

The diagram below shows a cross-section view of a duct with the locations of the inlet branches and exhaust pipe. A recommendation of hole size and spacing arrangement is shown in Table 4-2.



Legend	
A	Sampling pipe
B	Exhaust pipe
C	Air flow

Figure 4-22: Cross section view of pipe and hole setup for large duct

For large ducts, holes are nominally spaced each 400 mm (16 in).

Table 4-2: Hole size for large duct

Duct width	No. of holes	Hole Ø	Nominal pipe flow rate (L/min.) (cfm)
1 m (3 ft 4 in)	6	3.5 mm (9/64 in)	42.2 L/min. (1.49 cfm)
1.5 m (5 ft)	8	3.0 mm (1/8 in)	41.4 L/min. (1.46 cfm)
2 m (6 ft 6 in)	10	3.0 mm (1/8 in)	50.0 L/min. (1.76 cfm)

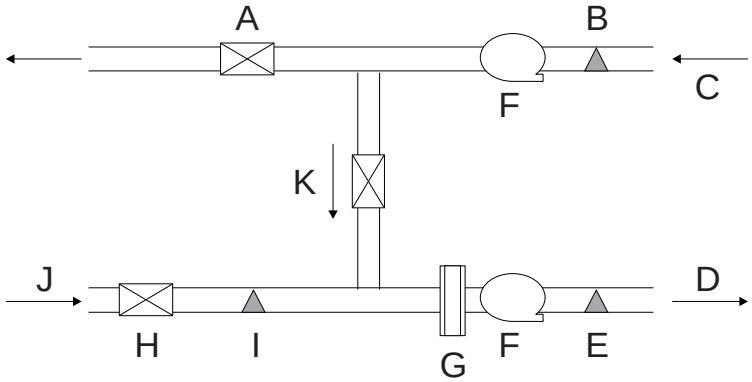
ASPIRE calculations shown in Table 4-1 and Table 4-2 apply to a 5 m (16.4 ft) inlet pipe and a 2 m (6.56 ft) exhaust pipe. Always check with local codes and standards for sampling hole spacing.

4.6.1 Condensation from Ducts

Condensation may occur when the air being sampled is warmer than the air surrounding the detector. Refer to the VESDA Pipe Network Installation Guide for information on how to avoid condensation problems.

4.6.2 HVAC System

HVAC protection is implemented for both return side and supply side to protect the HVAC fan and filter. Additionally, detectors may be placed in the outdoor air supply in order to identify whether external pollutants are being drawn into the system from outside of the building.

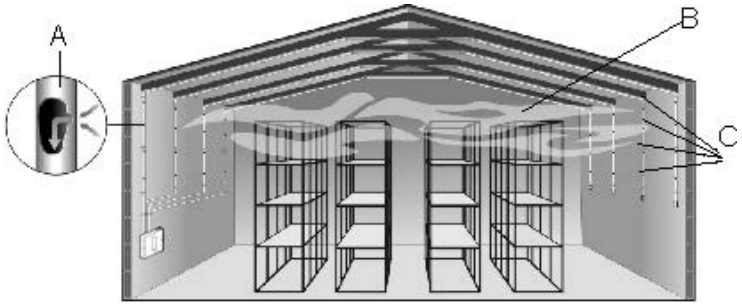


Legend	
A	Exhaust damper
B	Return air sampling
C	Return air
D	Supply air
E	Supply air sampling (fan, filter fire)
F	Fan
G	Filter bank
H	Supply air damper
I	Outside air monitor
J	Fresh air intake
K	Return air damper

Figure 4-23: HVAC system

4.7 Large Area Sampling

Areas such as atria and warehouses with high ceilings require special variations of the pipe network design rules. Stratification is a process where due to hot air layers closer to the ceiling, smoke loses its thermal energy, stops rising and levels out horizontally at a certain height. Stratification layers may be formed at different heights, restricting the smoke's ability to rise and reach the sampling pipe network. Factors such as temperature, ventilation, and roof height, all affect the degree of stratification and the level to which the smoke will rise. Changes in the above mentioned factors result in different stratification effects on a site. Where stratification is likely to occur, conventionally designed pipe networks may not be effective.



Legend	
A	Detail of sampling hole
B	Stratified smoke layer
C	Vertical sampling holes

Figure 4-24: Sampling air from areas with high ceilings

To overcome the stratification effect, a vertical sampling pipe may be installed in addition to the standard pipe on the ceiling. The vertical sampling pipe penetrates the stratification layers at different heights and samples the air at multiple levels. Follow local codes and standards for dimensioning. Heights exceeding value specified by local codes and standards should adhere to a performance-based design approach.

4.8 Dilution

If smoke concentrations occur at the extreme end of the sample pipe run, a smoke-laden air sample must pass a number of other sampling points, all of which are supplying only slightly polluted samples (if they are near the fire source) or completely clean samples (if they are far away from the fire source). The net result is the dilution of the smoke sample, leading to a lower reading after a longer time than would otherwise be the case.

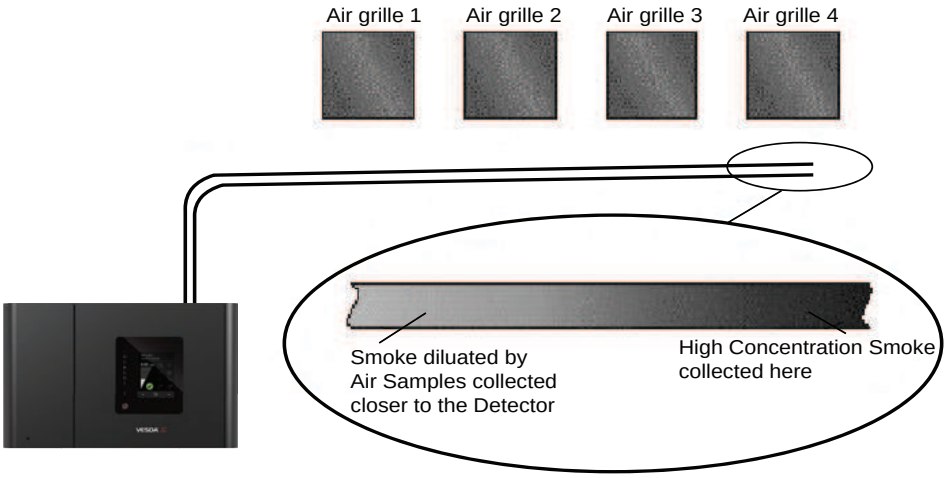


Figure 4-25: Smoke dilution effect over several air sampling points

4.9 Cumulative Sampling

Where HVAC systems distribute air within a protected area, any smoke will be dispersed and diluted. Traditional point detectors will wait until one detector reaches 100% of the smoke threshold before alarming. VESDA systems are able to **aggregate** the detected smoke from all of the sample points and can provide much earlier warning. For example, if you have a room with four sampling points and each detects 25% of the allowable smoke level the VESDA systems can aggregate the levels and alarm.

Cumulative sampling is a significant advantage in high air flow areas.

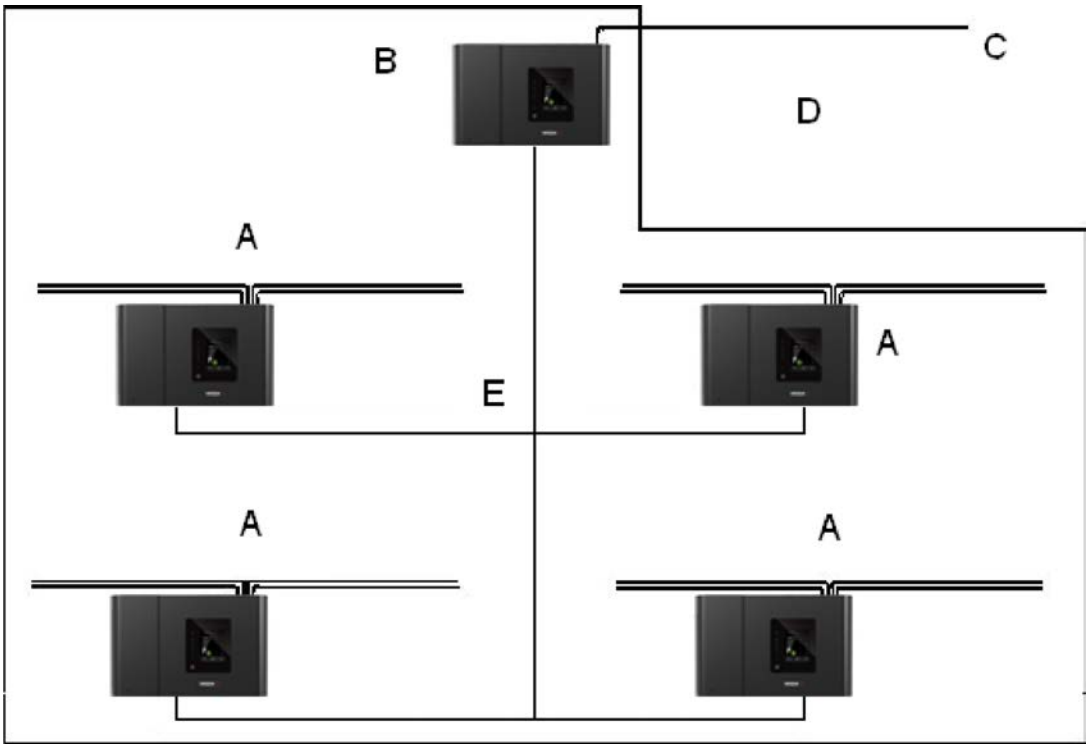
4.10 Reference Sampling

Periodically, smoke and other pollutants from external sources may enter a protected zone temporarily raising the smoke level. In this case, the detector will detect the smoke and generate an alarm. Referencing is employed to compensate for the periodic rise in smoke levels and to reduce nuisance alarms in high sensitivity areas.

A separate VESDA detector is used to draw air from the external source and produce a reference reading of the background level of smoke and pollutants. The reference reading is then subtracted from all VESDA detectors monitoring the internal VESDA Addresses. This allows the internal VESDA detectors to determine if a rise in smoke levels is due to background pollution or a problem inside the VESDA zone of protection. The ability to check the background level of smoke greatly reduces the chance of detectors false alarming.

VESDA detectors in large installations typically operate at varying sensitivity levels. In such instances the level of subtraction can be set differently for each detector.

Referencing is configured through Xtralis VSC, LCD Programmer or Xtralis VSM4 Software. Refer to the respective manuals for details.



Legend			
A	Internal detectors	D	Reference area (car park, loading bay, etc.)
B	Reference detector	E	VESDAnet
C	Reference sample pipe		

Figure 4-26: Reference sampling

4.11 Sampling Methods for Different Applications

The table below outlines different sampling methods for some of the most common applications:

Table 4-3: Sampling Methods for Different Applications

	Standard Sampling			Capillary Sampling			Return Air	
	Below Ceiling	In-Ceiling /Floor	Above Cabinet	Concealed	In Cabinet	Drop Pipe	In Duct	Return Air Grille
Clean Rooms	●	●					●	●
Computer Rooms	●	●	●		●		●	●
Electronic Data Processing (EDP) Environments	●	●	●		●	●	●	●
Containment Aisles								●
Control Rooms	●	●	●		●		●	●
Telecommunications	●		●		●			●
Theaters	●			●			●	●
Libraries/Archival Storage	●			●			●	●
Hotels				●			●	●
Hospitals	●	●		●			●	●
Casinos	●			●			●	●
Elec/Switching Cabinets			●		●	●		
Substations	●	●	●		●	●	●	●
Museums/Art Galleries				●			●	●
Offices	●	●		●			●	●
Historic Buildings				●				
Auditoria	●			●			●	●
Refrigerated Storage	●						●	●
Schools	●			●				
Storage Areas	●							
Equipment Cubicles			●		●	●		
Aircraft Hangars	●							
Atria	●			●		●	●	●
Cable Tunnels/Trays	●					●		
Dormitories				●			●	●
Laboratories	●	●			●		●	●
Manufacturing Facilities	●		●		●		●	●
Prisons				●			●	
Transportation				●				●

Note: For more vertical applications refer to appendix A.

4.12 Choice of Detector

The appropriate VESDA detector must be identified once site conditions have been assessed and the sampling method has been selected. Detector selection should be based on the required area coverage and the chosen sampling method. The table below identifies the suggested VESDA detector for different environments. Final detector selection, however, should be determined by actual site conditions and the specific pipe network design.

Table 4-4: Detectors suggested for different applications

	Detector	VLP	VLF	VLI	VEP-1	VEP-4	VES
Clean	Clean Rooms	•			•	•	•
	Computer Rooms/Data Centers	•	•		•	•	•
	EDP Environments	•	•		•	•	•
	Containment Aisles	•	•		•	•	•
	Control Rooms	•	•	O	•	•	•
	Telecommunications	•	•		•	•	•
	Theatres	•				•	•
	Libraries/Archival Storage	•		O	•	•	•
	Hotels	•	•			•	•
	Hospitals	•				•	•
	Casinos	•				•	•
	Elect/Switching Cabinets		•	O	•		
	Equipment Cubicles		•		•		
	Substations		•	•	•		
	Museums/Art Galleries	•				•	•
	Offices	•	•		•	•	•
	Historic Buildings	•	•	O	•	•	•
	Cold Rooms	•		•	•	•	•
	Auditoriums	•				•	•
	Refrigerated Storage	•		•	•	•	
	Schools	•			•	•	•
	Storage Areas	•		•	•	•	•
	Dormitories	•		•	•	•	•
	Atria	•				•	•
	Aircraft Hangars	•	•	•	•	•	•
	Laboratories	•			•	•	•
	Cable Tunnels/Trays	•		•	•	•	•
	Transportation	•	•	•	•	•	
	Prisons		•	O		•	•
	Manufacturing Facilities	•		•	•	•	•
	Warehousing	•		•	•	•	•
	Laundry	•		•	•	•	
	Sugar Mills			•			
	Textile			•			
	Processing Plants			•			
	Timber Milling			•			
	Water Treatment			•			
	Pulp and Paper			•			
	Petrochemical Plants			•			
	Fertilizer			•			
	Abattoirs			•			
	Conveyor	•		•		•	
	Mining	•		•		•	

Notes:

- Some applications identified as clean in Table 4-4, may include areas that are contaminated or considered dirty. Whilst this ‘Detector’ selection table provides guidance on the most appropriate detector selection for the application or environment, it should be used as a reference only.
- Areas such as roof spaces, electrical switches or substations in any application within the above list often have high levels of contamination that can affect standard detector performance. In such cases, the VESDA VLI may be more suitable. Some applications are indicated by the symbol “O“.
- Table 4-4 does not include all applications for which VESDA is suitable or can be applied.
- VESDA-E VEU can be used in applications to provide the highest smoke sensitivity, longer pipe run, and larger coverage area.

4.13 Detector Parameters

Different detectors in the VESDA range have different characteristics. The pipe network design parameters for each of these detectors may vary according to site conditions and requirements. The parameters for VESDA detectors are shown below:

Table 4-5: VESDA detector parameters

Parameters	VLF-250	VLF-500	VESDA VLP*	VESDA VLI	VESDA-E VES	VESDA-E VEP		VESDA-E VEU
						VEP-1	VEP-4	
ASPIRE support	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Area Coverage	250 m² (2,690 sq.ft)	500 m² (5,380 sq.ft)	2,000 m² (21,520 sq.ft)	2,000 m² (21,520 sq.ft)	2,000 m² (21,520 sq ft)	1,000 m² (10,760 sq. ft)	2,000 m² (21,520 sq ft)	6,500 m² (69,965 sq ft)
Pipe Length (Linear)	25 m (82 ft)	50m (164 ft)	400 m (1,312)	360 m (1,181 ft)	280 m (919 ft)	100 m (328 ft)	280 m (919 ft)	400 m (1,312 ft)
Pipe Length (Branched)	30 m (98 ft)	60m (197 ft)	640 (2,100 ft)	445 m (1,460 ft)	560 m (1,837 ft)	130 m (427 ft)	560 m (1,837 ft)	800 m (2,625 ft)
No. of Pipe Inlets	1	1	4	4	4	1	4	4
Maximum length for each pipe	N/A	N/A	100 m (328 ft)	90 m (295 ft)	70 m (230 ft)	100 m (328 ft)	70 m (230 ft)	100 m (328 ft)
Maximum branch length	15 m (49 ft)	30m (98 ft)	N/A	N/A	N/A	65 m (213 ft)	N/A	N/A
Maximum sampling holes per pipe (including end cap hole)	12	30	25	24/28/60	40/80/100	30/40/45	40/80/100	80/80/100
Maximum length of capillaries	4 m (13 ft) for small bore (5.2 mm) tube and 8 m (26 ft) drop pipes.							
Aspirator speed	Fixed	Fixed	Variable	Variable	Variable	Fixed	Variable	Variable
Minimum system air flow	12 liters/minute	12 liters/minute	20 liters/minute	40 liters/minute	40 liters/minute	40 liters/minute	40 liters/minute	40 liters/minute
Minimum air flow per pipe	N/A	N/A	15 liters/minute	20 liters/minute	20 liters/minute	N/A	20 liters/minute	20 liters/minute
Minimum sector pressure	65 Pa	65 Pa	70 Pa	70 Pa	70 Pa	70 Pa	70 Pa	70 Pa

* Only available in China.

The parameters listed above are for a typical pipe network. Site conditions and requirements will dictate the final parameters for each site. The aggregate pipe length can exceed the published values, provided the minimum pressure at each sampling point is at least 25 pascals. It is recommended that transport time for the network is maintained at 60 seconds or less and hole balance and hole share be at least 70% (use ASPIRE to calculate these values). These values may be relaxed subject to local fire codes and conditions. Exceeding the guidelines specified in this guide is acceptable, provided the design is validated and confirmed by the ASPIRE software.

These pipe lengths represent best practice for systems with single pipe runs on each port (no branching). Longer and/or more complex pipe arrangements are also possible and must be verified using ASPIRE to ensure transport time and system balance requirements are met.

4.14 Additional Monitoring Equipment

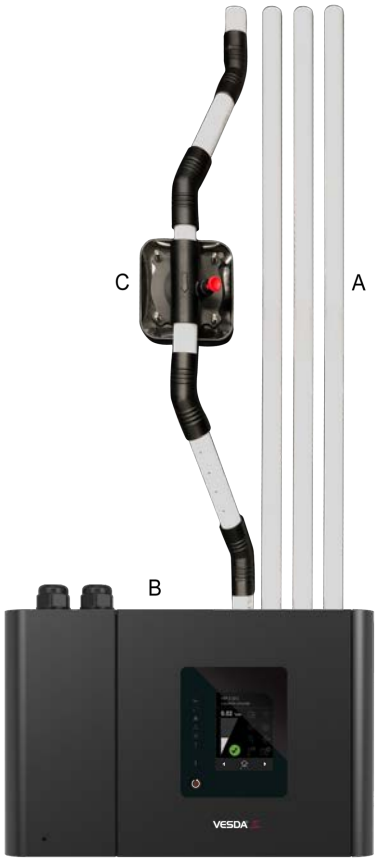
In many environments, protection is required not only against fire, but also to protect personnel against dangerous environments.

4.14.1 Gas Detection

For occupational health and safety reasons, there is a need to monitor for dangerous gases in many environments.

It is possible to utilize the VESDA air sampling technology by incorporating VESDA Sensepoint XCL - Large Bore to detect the presence of these gases in a protected environment. This turnkey solution provides many benefits including cost effectiveness, ease of maintenance and management.

Refer to the VESDA Sensepoint XCL - Large Bore Installation manual (Doc. No. 35563) for further information.



Legend	
A	Sampling Pipes
B	Detector
C	VESDA Sensepoint XCL - Large Bore

Figure 4-27: VESDA Sensepoint XCL - Large Bore gas detection equipment

5 Defining the Site

5.1 Regulatory Requirements

Local codes and standards determine the maximum spacing between pipes and sampling holes. These maximums may change depending on the environment being protected. Local codes and standards for aspirating smoke detectors will have precedence over any parameters suggested by Xtralis. Some of the key requirements are listed below.

- Maximum permissible transport time
- Maximum area for a fire zone
- Area of coverage per sample hole (point detector)
- Maximum spacing between sampling points
- Maximum area for aspirating systems

5.2 Fire Zones and VESDA Addresses

Fire zones are created to meet regulatory requirements, whereas, VESDA Addresses are areas that can be monitored by one VESDA detector. VESDA addresses are created for the system to operate within defined parameters for optimum smoke detection.

Site conditions have an impact in defining VESDA Addresses. Some key guidelines need to be observed when creating an Address.

- One detector can monitor only one VESDA Address
- The VESDA Addresses must comply with the local codes and standards
- The environmental conditions within each VESDA address should be the same. For example, the level of pollution and ambient air pressure within each VESDA Address should not change.
- The appropriate VESDA detector parameters are met.

Legend					
A	Single physical fire zone	C	CPUs and storage sub-systems	E	Power supply
B	Two VESDA Addresses	D	Modems, printers and multiplexers	F	Media storage

Figure 5-1: Constructing VESDA Addresses

6 Planning and Mapping Pipe Network

In planning a pipe network, you must:

- Have an understanding of the area to be protected
- Understand the environmental conditions of the protected area
- Identify any forced air ventilation and air flows (fans, air conditioners, etc.)
- Take into account any customer specifications
- Determine if the design needs to address:
  - Return air grille sampling
  - Ceiling and under floor voids
  - Consideration for high ceilings, stratification effect and containment aisles
  - Any requirement for focused detection

The pipe network grid is mapped on to the construction drawings. The objective of mapping the pipe network is to determine the placement of sampling holes and to optimize the location of the VESDA detector with a view to minimizing the pipe length. While mapping the pipe network, care should be taken to minimize the number of bends and elbows used. We recommend that multiple pipes are used in preference to changes in pipe direction. Refer to Figure 9-1 on page 39.

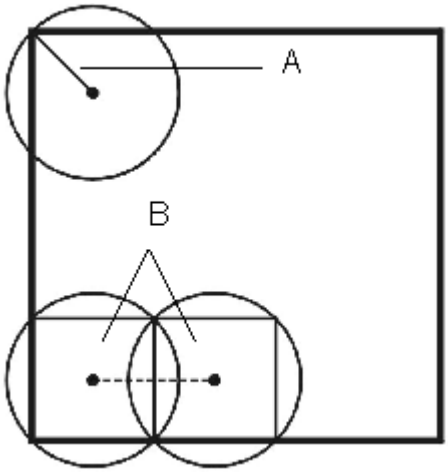
6.1 Grid Overlay

The Grid Overlay method is utilized to map the pipe layout and determine the position of sampling holes. The dimensions of the grid depend upon the required maximum or minimum sampling point separations and the required distance of sampling holes from the walls, which are usually defined by local codes and standards.

Ideally the aim should be a square grid, however, the measurements and the shape of the area to be covered will determine the grid.

Plot the first sampling hole in a manner that it does not exceed the maximum spacing as dictated by local codes and standards, or by installation requirements. Typically the maximum distance is likely to be to the corner of the room.

Note: The spacings imposed by your local codes and standards may well be related to the cost of conventional point detectors. With the VESDA system, it is possible to substantially increase the density of sampling points at negligible cost. Grids of 4 m X 4 m (13 ft X 13 ft), 6 m X 6 m (20 ft X 20 ft), or 4 m X 8 m (13 ft X 26 ft) are popular choices.



Legend	
A	Maximum distance of sampling hole from the wall. (Max 5.1m or 16ft)
B	Spacing between sampling holes (Max 10.2m or 32ft)

Figure 6-1: Illustration of plotting sampling holes and spacing between holes

6.1.1 Sampling Hole Conventions

The below example relates to NFPA 72 requirements:

- The typical maximum size for a fire zone is 2000 m² (21520 sq.ft)
- The maximum area covered by a sample point is 82.81 m² (891 sq.ft)
- The sample hole will not be more than 4.55 m from walls (15 ft)
- The maximum space between sampling holes is 9.1 meters (30 ft)
- A sample point shall not be more than 6.4 meters (21 ft) from any point in the room

Plot the remaining sampling holes for the row. Normally these will be spaced at equal distance from each other.

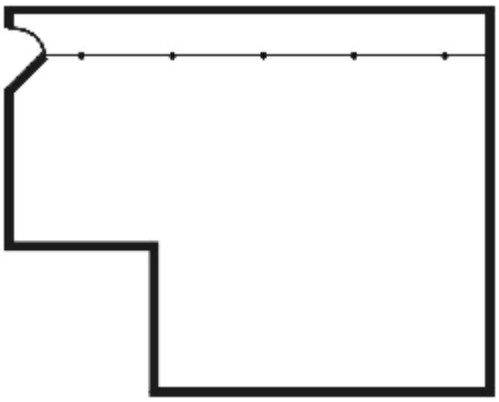
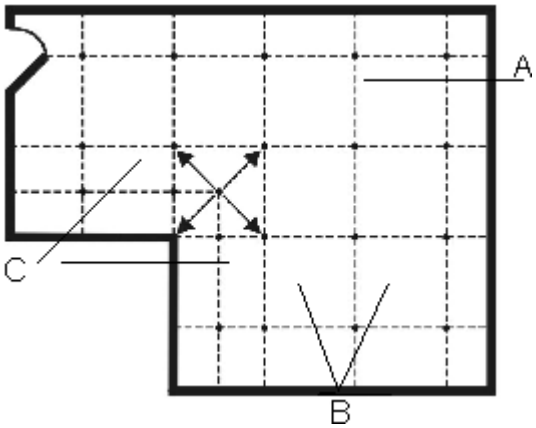


Figure 6-2: Plotting the first row of sampling holes

Plot sampling holes in equidistant parallel rows to form squares. If the area to be plotted is irregularly shaped a combination of square and/or rectangular plotting may be required.



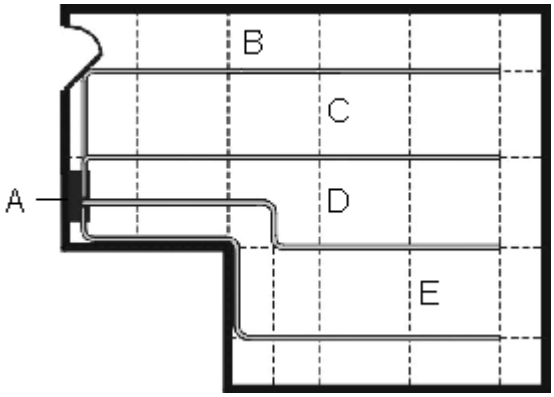
Legend	
A	Square plotting of sampling holes
B	Grid overlay
C	Rectangular plotting of sampling holes

Figure 6-3: Square and rectangular plotting of sampling holes

After plotting the sampling holes, determine the optimum positioning of the VESDA detector, taking into account the following good design principles:

- Shorter multiple pipe runs. To find out why this is critical, refer to the System Performance Graph in Figure 9-2 on page 40.
- Minimum changes in pipe direction

Next, plot the pipes on to the site plan by joining the holes and terminating these at the detector.



Legend	
A	Detector
B	Sampling pipe 1
C	Sampling pipe 2
D	Sampling pipe 3
E	Sampling pipe 4

Figure 6-4: Plotting the sampling pipes by connecting the sampling holes and terminating at the detector

6.2 Site Parameters

Each site presents its own unique set of parameters which the pipe network must meet. Some factors likely to influence pipe network design are:

- Level of protection required
- The area to be covered
- The environmental conditions
- The layout (of plant, machinery, equipment or furniture)
- Airflows
- External influences
- Special equipment to be protected (refer to Section 7.3)
- Combustibility of material (construction and stored)

6.3 Client Parameters

The client may specify certain requirements to be included into the design. These may typically relate to protection of certain equipment or the threshold levels for alarms. The pipe network design should incorporate such specifications subject to regulatory and detector parameters.

6.4 Performance-Based Parameters

Performance-based design provides an alternate fire protection system to prescriptive fire codes. They do this by assessing the environmental risks at the concept design stage. This design approach offers significant advantages. The most important one is the ability to provide early detection of a fire event.

It is recommended that smoke testing or Computational Fluid Dynamics (CFD) modelling be performed. This method of airflow simulation is used to determine the optimal location for the VESDA system by accurately identifying smoke travel from previously acknowledged risks.

In areas where performance-based design is not recognized, its concepts may still be adhered to by incorporating the design basics from prescriptive codes (i.e. NFPA 318 and SEMI s14 - 2000), and the basics outlined in this guide.

The arrangement of process tools and equipment will alter the airflow dynamics (air speed and air direction) in the facility. To ensure that the system design is effective, it is recommended that performance tests are conducted during the completion stage of the site construction.

7 Designing Pipe Networks for Specific Applications

VESDA systems can be fine - tuned to most environments. Each environment presents its own special requirements and these are addressed during pipe network design. Table 4-3 on page 23 suggests air sampling methods normally used for different types of applications. A combination of different sampling methods may be used to create the optimum protection for a site. Four basic applications are discussed below:

- Protecting standard rooms
- Protecting high air movement areas
- Localized detection
- High ceiling areas

7.1 Standard Rooms

The typical standard room pipe network is an on ceiling sampling method as described in Chapter 2.

7.2 High Air Exchange Rooms

High air exchange areas normally use some form of mechanical ventilation to maintain dust free operations or provide for heat dissipation. Most local codes and standards support that any environment with 7.5 air changes per hour (ACH) or more is classified as a high air volume exchange area.

The sampling methods used for high air movement areas are a combination of return air and open area. The return air sampling may be conducted at a return grille or via in-duct sampling. In high air exchange environments, the air flow direction is influenced by artificial means. Any air volume outside the direct air flow path may not reach the sampling points on the return grilles.

Below ceiling sampling is used to sample air from areas that fall outside the direct air flow path created by AHUs, and for fire detection when the air-condition is off. Figure 7-1 on page 34 illustrates the use of return air sampling in combination with on ceiling air sampling in areas of high air exchange.

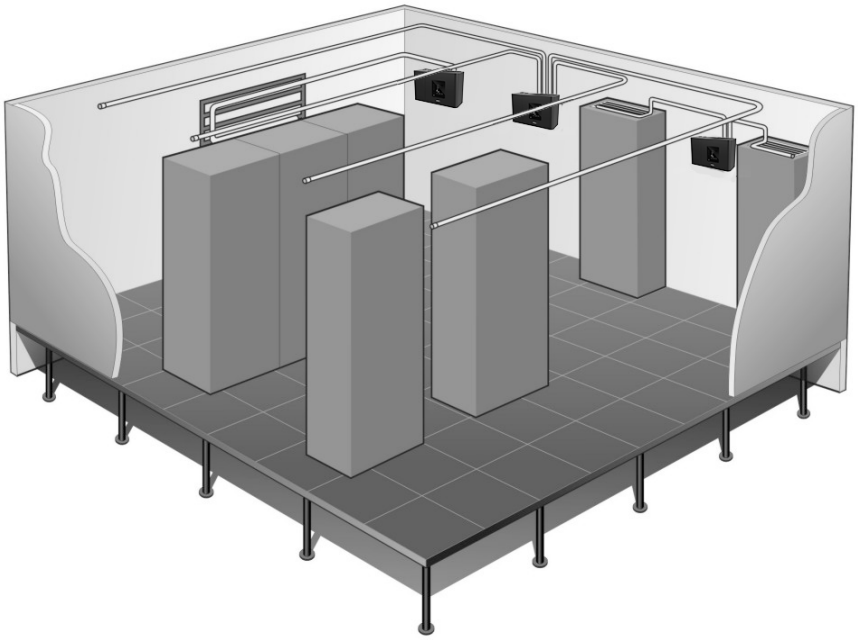
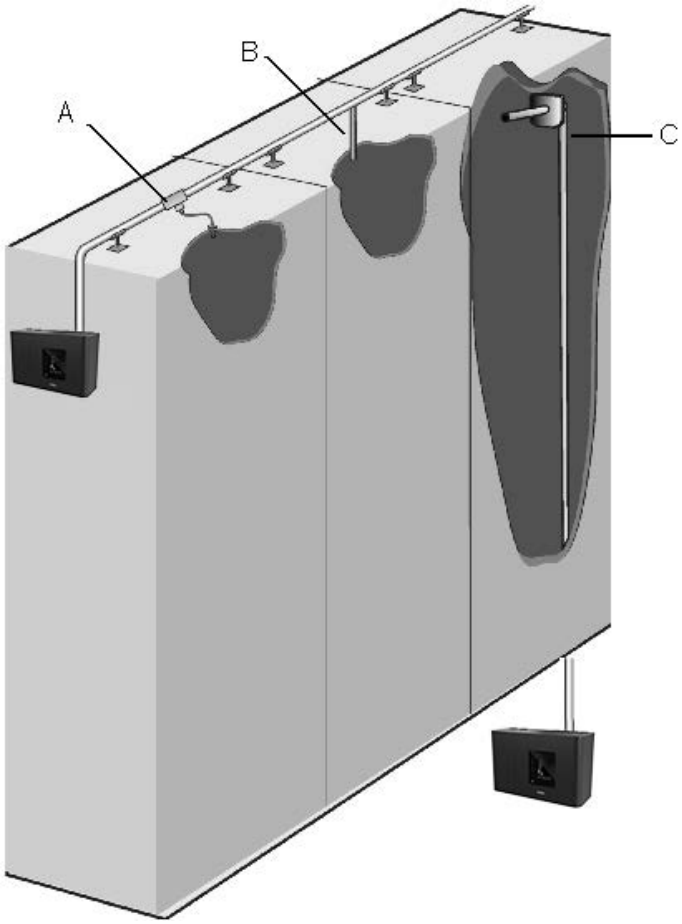


Figure 7-1: In-ceiling and return air sampling for rooms with high air exchange

Spacings for the return air and below ceiling samplings should follow the requirements of local codes and standards.

7.3 Localized Detection

There are instances where a specific object within a room requires special monitoring. In-cabinet sampling uses capillaries to sample air from inside cabinets which are used for electrical, IT, or other mission-critical purposes. The capillaries are suspended from a sampling pipe at the top of the equipment cabinet, or come up from an under floor void.



Legend	
A	Capillary
B	Drop pipe
C	Floor void capillary

Figure 7-2: In-cabinet air sampling with capillaries, drop pipes and under floor sampling methods

When installing in-cabinet sampling from under the floor, the capillary will still always be suspended inside the top of the cabinet.

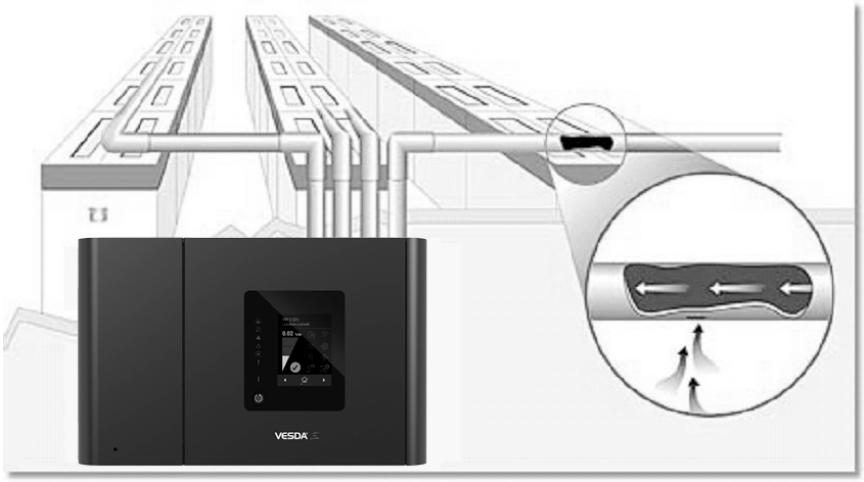
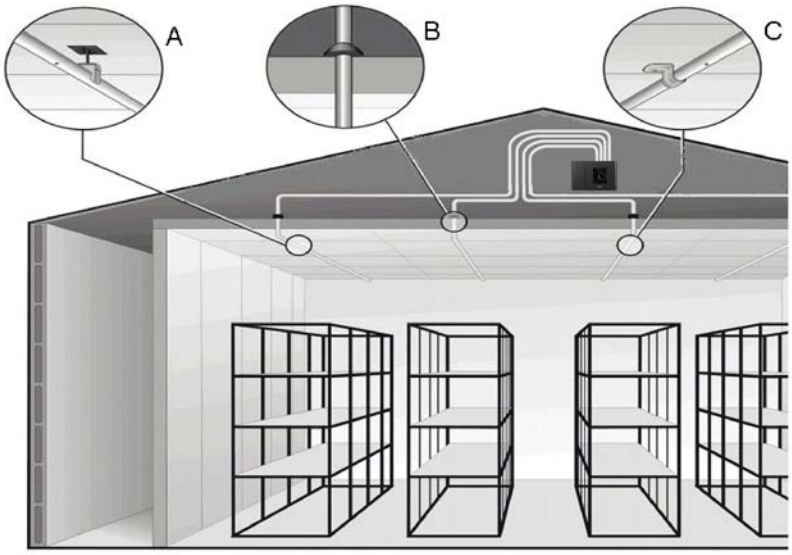


Figure 7-3: Above cabinet sampling

7.4 High Ceiling

VESDA Addresses with high ceilings require special consideration because of the stratification effect. To overcome the stratification effect, one or more vertical pipes are installed in addition to the pipes on the ceiling. Refer to Section 4.7 for an example of pipe network installation for areas with high ceilings.



Legend	
A	Pipe held off roof
B	Sealed entry to freezer
C	Normal pipe installation

Figure 7-4: Freezer warehouse installation

For further information on designing refrigerated storage, refer to the VESDA Warehouse and Refrigerated Storage Design Guides.

8 Testing Design Performance

You can test the design performance with the ASPIRE software provided by Xtralis. The software is used to evaluate the performance differences that occur by making changes to the pipe design. For the latest version of the software, visit Xtralis website (www.xtralis.com) or contact your local Xtralis office.

The efficient performance of a pipe network design is dependent upon the ASPIRE software generating acceptable values for:

- Hole % balance
- Maximum transport time
- Hole sensitivity

Factors influencing these values are:

- Pipe length
- Number of sampling holes
- Size of the end cap hole
- Number and radius of bends
- Number of pipes
- Length of capillary tubes
- Size of sampling holes
- Aspirator speed

8.1 Hole Balance

Hole % balance is the amount of air being drawn from the sampling hole with the least amount of air flow, divided by, the average air flow through the other sampling holes.

Subject to local codes and standards, a minimum hole balance of 70% or greater is recommended.

The aim of an optimum pipe design is to have a relatively equal air flow through **all** sampling holes. To achieve this, the Hole % Balance should be as close to 100% as possible. Pipe length, number of sampling holes, hole size, and end cap hole size all affect the Hole % Balance.

Reducing the number of holes per pipe, and using multiple pipes, will increase the balance.

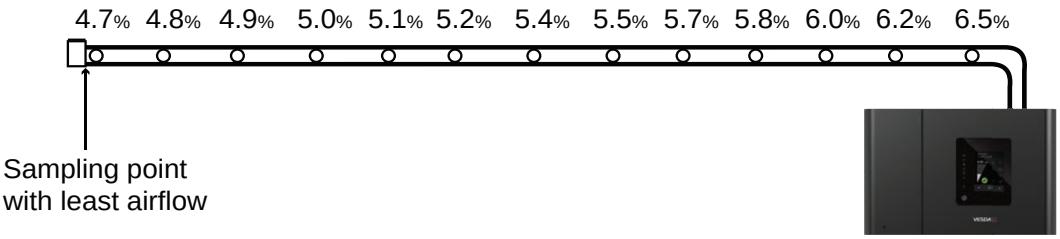


Figure 8-1: Hole balance

The mathematical formula is:

$$\text{Balance} = \frac{\text{Min Flow}}{\text{Max Flow}} \times 100 \qquad \text{Balance} = \frac{4.7\ \%}{6.5\ \%} \times 100 = 72.3\%$$

8.2 Maximum Transport Time

Maximum transport time indicates the longest time taken for smoke to travel from a sampling hole to the detector. You should aim to get the lowest possible time without compromising on the Hole % Balance and the Hole % Share values.

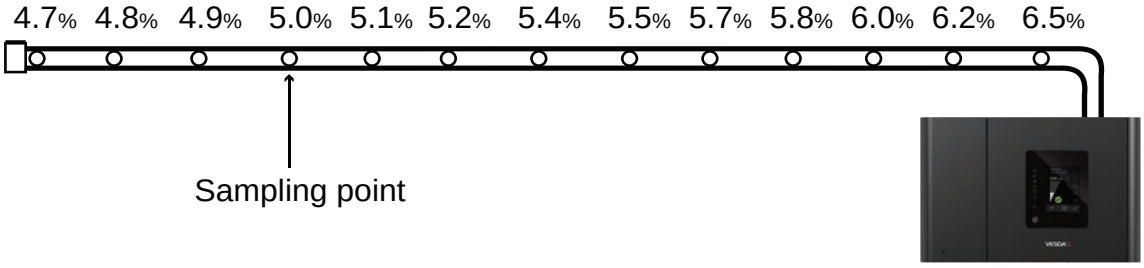
Note: Subject to local codes and standards the maximum transport time is 120 seconds. We recommend the maximum transport time of less than 60 seconds when early warning is desired.

Pipe lengths affect the time to transport a sample of air from the collection point to the detector. Factors like bends, diameter of sampling holes and end cap holes also affect the transport time. Figure 9-1 on page 39 illustrates how the same area can be covered by different pipe lengths.

8.3 Hole Sensitivity

The relative sample hole sensitivity is a measure of the sample hole in relation to the detector's fire threshold sensitivity. Mathematically it is presented as:

Detector Sensitivity / (Sample hole flow / Sum of all sample point flows) = % obs/m (%obs/ft)

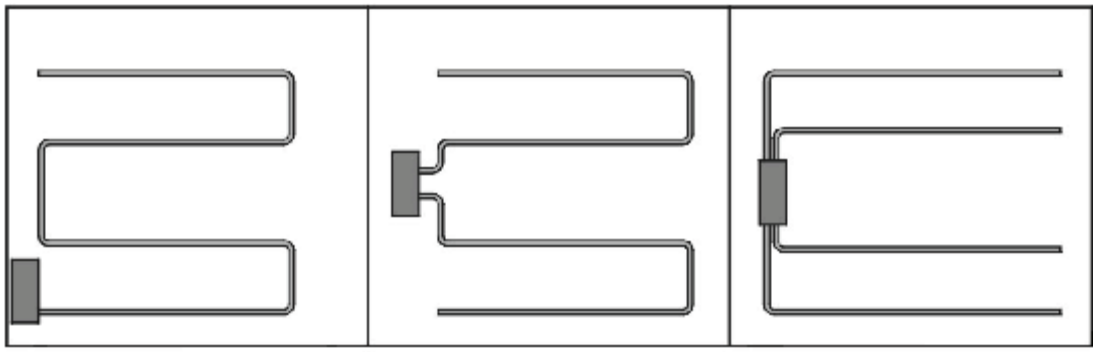


$$\frac{\text{Detector Sensitivity}}{\text{Sample Hole Flow / Sum of all Sample Point Flows}} = \frac{0.1}{0.05} = 2.0\%\ \text{Obs/m}$$

Figure 8-2: Hole sensitivity

9 Advantages of Multi Pipe Systems

Subject to the maximum length parameters for the detector it is possible to design a pipe network covering an entire zone with one pipe. However, for improved pipe network efficiency we strongly recommend you use shorter multiple pipes.



1 Pipe	2 Pipes	2 Pipes
100 m (328 ft)	50 m (164 ft) each	25 m (82 ft) each

Figure 9-1: Single and multiple pipe network systems for the same area

The advantages of multiple pipe systems are:

- Shorter pipes have better response times as the air transport distance from the end of pipe to the detector is reduced. Refer to Section 8.2 for details.
- Shorter pipes have less sampling holes per pipe and can have smaller end cap holes. This creates better hole share and hole balance. Refer to Section 8.1 for details.
- Shorter pipe lengths are likely to have fewer bends resulting in shorter transport times and better air flow
- System balance: multiple pipes provide better system balance

9.1 System Performance Graph

The system performance graph illustrates the advantages of using shorter, multiple pipes over fewer, longer pipe lengths. A single pipe of 100 meters has a transport time of 80 seconds. Two pipes of 50 meters have a transport time of around 33 seconds. Four pipes of 25 meters have a transport time of 20 seconds. Each of these configurations will protect the same area, but have substantially different transport time performance.

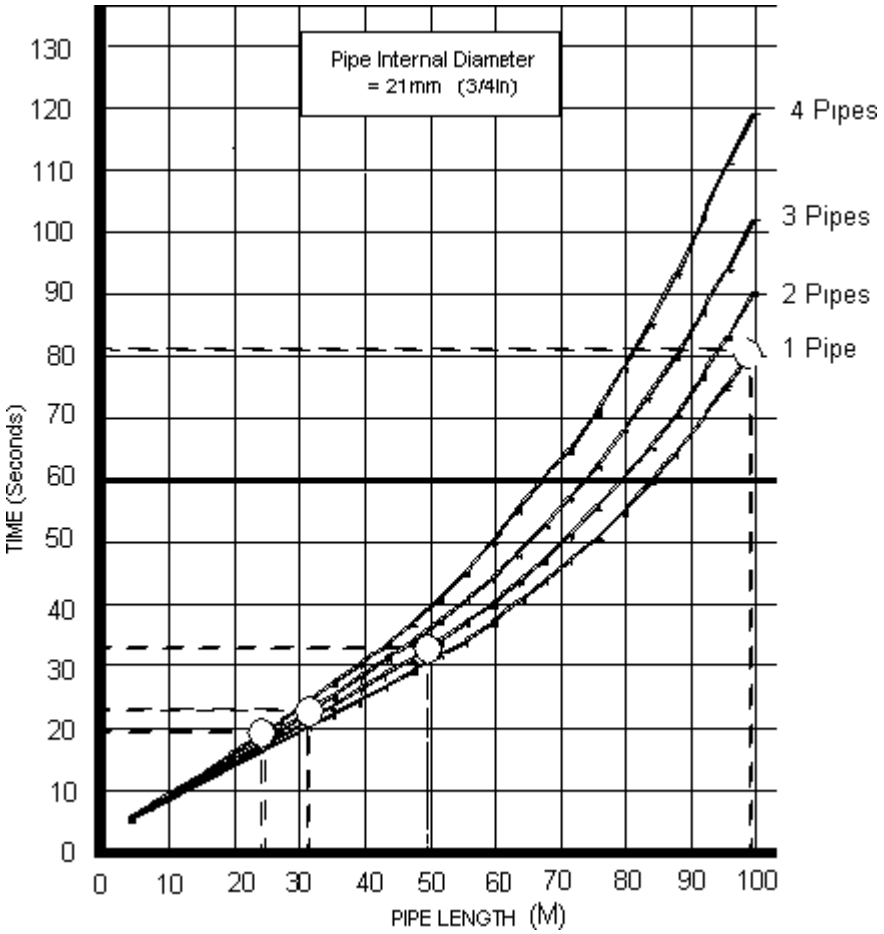


Figure 9-2: System performance graph

9.2 Sampling and End Cap Holes

The sampling holes are drilled into the pipe(s), once the pipe network has been installed. End caps must always be placed at the end of each sampling pipe. ASPIRE calculations will determine the diameter of each sampling hole and the end cap hole. The end cap hole can be used as another sampling hole, but is generally used to improve the transport time for the pipe. The diameter of the end cap hole is determined by the parameters of the pipe network, such as the pipe length and the number of sampling holes.

9.3 Pipe Connections

9.3.1 Joints

Pipe joints must be airtight. Glue together the pipes once the final form of the network has been determined. Do NOT glue the pipes to the pipe inlet and exhaust manifolds of the detector. The use of excessive amounts of glue can block or partially block pipe airflow to the detector and severely affect the detectors ability to detect smoke. Testing must be conducted after the pipes are glued together.

9.3.2 Bends

We recommend bends and elbows to be used to change the direction of the pipe. Wide radius bends are preferred to elbows. Good pipe network designs keep the number of bends and elbows to a minimum as they interfere with the optimum airflow. Figure 9-1 illustrates the advantage of multiple pipes over bends and elbows.

9.3.3 Tees

Tees are used for branching a sampling pipe or for attaching capillaries and drop pipes to the air sampling pipe.

10 Recording Pipe Network Design Specifications

Keeping an accurate record of the pipe network design specifications assists the installation engineer to correctly configure the pipe network. This information is also useful to complete the commissioning form. Information that needs to be recorded includes:

- Site name and address
- Site use (application)
- Site measurements and layout
- Factors requiring special attention
- The number and location of VESDA addresses
- Grid overlay showing the pipe network layout with sampling hole positions
- Actual sampling method(s)
- Actual VESDA detector
- ASPIRE results
- Actual alarm thresholds

A Appendix I. Applications

Education

- Atriums
- Dormitories/ Housing
- IT/Data processing area
- Laboratories
- Lecture, Theatres
- Maintenance Facilities
- Admin/Office Spaces

Data Centres/Telecom

- Battery Room
- Cable Vaults/Tunnels
- Generator Room
- Computer Room Air Handler (CRAH) unit
- Computer Room Air Conditioner (CRAC) unit
- Server Hall
- Service rooms (Mechanical, Electrical, Chiller Plant)

Industrial

- Cable Vaults
- Control Rooms
- Filter Banks
- Service Tunnel
- Service Rooms
- Manufacturing/ Service areas

Healthcare

- Atrium
- Elevator/ Lift
- IT/Data processing area
- Laboratories
- Medical storage
- MRI/Imaging Rooms
- Patient wards
- Service room (Mechanical, Electrical, Chiller Plant, Laundry)

Correctional

- Admin/Office Spaces
- Cells
- Control Units
- Infirmary
- Service room (Mechanical, Electrical, Library, Laundry)
- Supply/Return Ducts

Retail/Commercial

- Atriums
- Duct Detection
- Loading Bays
- Car Parks
- Retail Areas
- Service Rooms

Power Generation

- Battery Rooms
- Service Tunnels

- Control Rooms
- Generator Halls
- IT/Data processing area
- Switch Rooms
- Wind Turbines

Transportation

- Airplane Hangars
- Baggage Handling
- Passenger Terminals
- Train Tunnels
- Road tunnels

Warehousing/Logistics

- Admin/Office Spaces
- Cold-Storage
- High-Bay Storage
- In-Rack Detection
- Vehicle Charging

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